Topic 6 – Microbiology, Immunity and Forensics

Candidates will be assessed on their ability to:

6.1	understand the principles and techniques involved in culturing microorganisms, using aseptic technique
6.2	understand the different methods of measuring the growth of microorganisms, as illustrated by cell counts, dilution plating, mass and optical methods (turbidity)
6.3	understand the different phases of a bacterial growth curve (lag phase, exponential phase, stationary phase and death phase) and be able to calculate exponential growth rate constants
6.4	**CORE PRACTIAL 13** **Investigate the rate of growth of microorganisms in a liquid culture, taking into account the safe and ethical use of organisms.**
6.5	(i) be able to compare the structure of bacteria and viruses (nucleic acid, capsid structure and envelope) with reference to Ebola virus, tobacco mosaic virus (TMV), human immunodeficiency virus (HIV) and lambda phage (λ phage) (ii) understand what is meant by the terms *lytic* and *latency*
6.6	understand how *Mycobacterium tuberculosis* and human immunodeficiency virus (HIV) infect human cells, causing symptoms that may result in death
6.7	(i) know the major routes pathogens may take when entering the body (ii) understand the role of barriers in protecting the body from infection, including skin, stomach acid, and gut and skin flora
6.8	understand the non-specific responses of the body to infection, including inflammation, lysozyme action, interferon and phagocytosis
6.9	understand the roles of antigens and antibodies in the body's immune response including the involvement of plasma cells, macrophages and antigen-presenting cells
6.10	understand the differences between the roles of B cells (B memory and B effector cells), and T cells (T helper, T killer and T memory cells) in the host's immune response
6.11	understand how individuals may develop immunity (natural, artificial, active and passive)
6.12	understand how the theory of an 'evolutionary race' between pathogens and their hosts is supported by evasion mechanisms shown by pathogens
6.13	understand the difference between bacteriostatic and bactericidal antibiotics
6.14	**CORE PRACTICAL 14** **Investigate the effect of different antibiotics on bacteria.**
6.15	know how an understanding of the contributory causes of hospital-acquired infections has led to codes of practice regarding antibiotic prescription and hospital practice that relate to infection prevention and control

6.16	know the role of microorganisms in the decomposition of organic matter and the recycling of carbon
6.17	know how DNA can be amplified using the polymerase chain reaction (PCR)
6.18	know how gel electrophoresis can be used to separate DNA fragments of different length
6.19	understand how DNA profiling is used for identification and determining genetic relationships between organisms (plants and animals)
6.20	understand how to determine the time of death of a mammal by examining the extent of decomposition, stage of succession, forensic entomology, body temperature and degree of muscle contraction